

AI-POWERED SURFACE DEFECT DETECTION

Catch every strip defect at the line, not at the customer

A 6.22 MB computer-vision model that detects, classifies and scores surface defects in real time, calibrated to what a wrong call costs the mill, and running live today.

TEAM

[TEAM NAME]

[Member 1] · [Member 2] · [Member 3] · [Member 4]

[INSTITUTION]

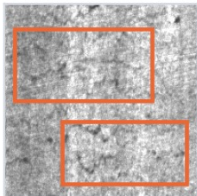
TRY THE LIVE DEMO



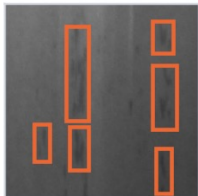
surface-vision.github.io

The trained detector, running in your browser. Upload a strip image; nothing leaves the page.

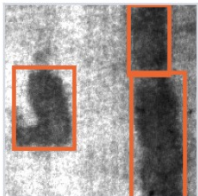
crazing



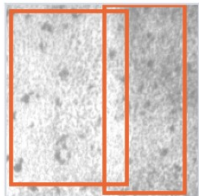
inclusion



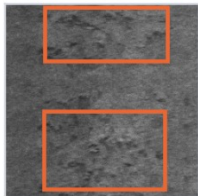
patches



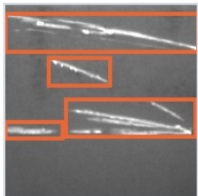
pitted surface



rolled-in scale



scratches



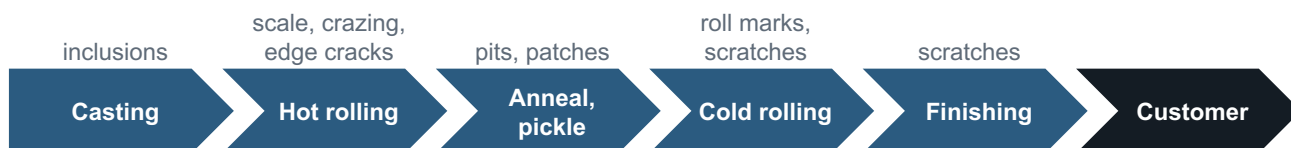
Six defect families shown on held-out test frames.

Slide 1: problem and solution. Slide 2: validation, roadmap and value.

Why is late detection costly, and what did we build to fix it?

Cost of Late Detection

research_notes s2a, s4a

**AI target:** classify each defect where it is born**Found today:** by eye, or by the customer**8-10%**

nickel in grade 304 by mass, so a stainless downgrade costs more

300 m/min

strip speed at which the human eye cannot inspect the full surface

~15%

of sales lost to poor quality in manufacturing, range 5-35%

Model Options Tested

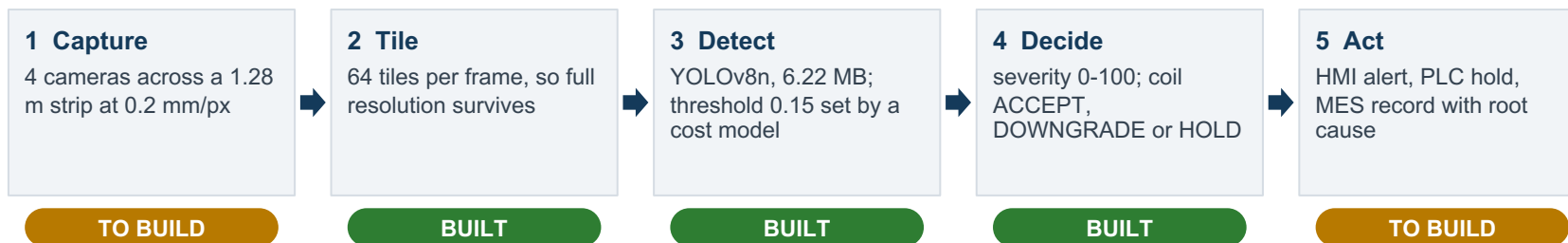
model_study, gap1 reports

OPTION	mAP50	COST	VERDICT
YOLOv8n at 256 px	0.752	3.0 M params	CHOSEN
YOLOv8n + Severstal clean strip	0.764	same size	READY
YOLOv8s, 3.7x the parameters	0.734	3.5x FLOPs	NO GAIN
YOLOv8n retrained at 640 px	0.734	4x pixels	NO GAIN
Test-time augmentation	-0.006	2.39x latency	REJECTED

Held-out test, scored once. Nano vs small: 95% CI [-0.008, +0.042] spans zero.

Solution Architecture

src/inference.py, src/report.py



EVERY DEFECT RETURNS

Bounding box

Defect type

Calibrated confidence

Severity 0-100

Root cause and fix

Closed loop (proposed): operator overrides feed active learning; a per-line retrain takes 0.6 h.

Live Demo

checked 13 Sep 2026

surface-vision.github.io

Upload a strip image
Box, type, confidence, severity
30-50 ms per frame on CPU
No image leaves the browser

Late, sampled inspection is the cost. A 6 MB detector that classifies every defect at line speed and names its process owner is the fix, and it already runs live.

Does it meet the brief, and what should Jindal do first?

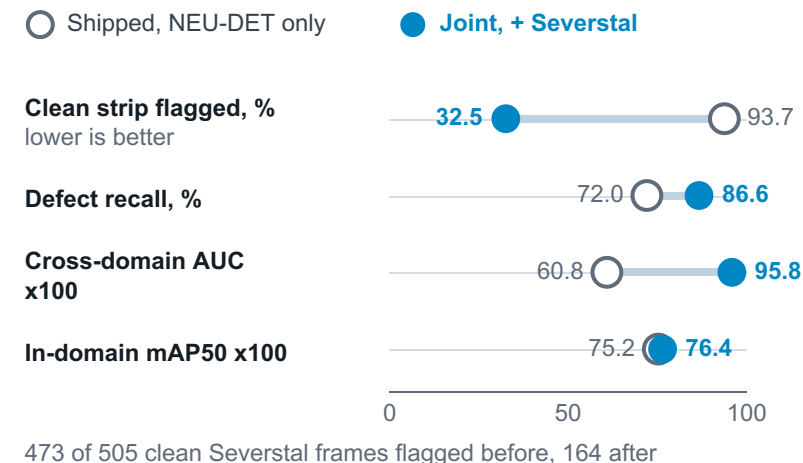
Scorecard Against the Brief

reports/*.json, exec_summary_checks.json

CONSIDERATION	EVIDENCE, HELD-OUT DATA	STATUS	NEXT STEP
Detection accuracy	mAP50 0.752 on held-out test; all 180 defective frames flagged	MET	-
False-alarm rate	Clean strip flagged 93.7% -> 32.5% after joint training; confidence calibrated, ECE 0.142 -> 0.046	PARTIAL	Jindal clean strip, P1
Multiple defect types	Type right on 177/180 frames. AP50: scratches 0.91, scale 0.63, roll marks 0.20; no public edge-crack data	PARTIAL	Plant labelling, P1
Line-speed inference	6.22 MB, 30-50 ms per frame on CPU. At 250 m/min: 18 accelerators at 0.2 mm/px, or 1 downsampled	MET	Resolution spec, P0
Usable demo	Live web demo: upload an image, get box, type, confidence, severity and root cause	MET	-
Scaling across grades	Second dataset: cross-domain AUC 0.608 -> 0.958 and in-domain mAP50 up to 0.764; retrain 0.6 h	PARTIAL	Stainless grades, P1-P4

Training on Real Clean Strip

gap1 reports, conf 0.15



Pilot Roadmap, 20 Months

phase plan; 0.6 h retrain measured

PHASE	m0	m4	m8	m12	m16	m20	EXIT GATE
P0 Instrument one line	■		pilot go / no-go				Frames captured at line speed
P1 Fine-tune on Jindal strip		■					Clean-strip alarms under target
P2 Shadow mode, no interlock			■				Operator-verified alarm rate
P3 PLC hold on severe defects				■			Hold decisions trusted
P4 Second line and grade					■		Fleet rollout decision

Business Case

research_notes s3a, FY26

2.57 Mt x 1% downgraded x 10% discount x INR 167,407/t
= INR 43 cr a year at risk

Downgraded \ discount	10%	15%	20%
0.5% of volume	21	32	43
1.0% of volume	43	64	86
2.0% of volume	86	129	172

Recovering 30% is about INR 13 cr a year: payback 0.7-2 years on a USD 1-3M line system (unverified range).

Needed from Jindal: the real downgrade rate.

Recommendation: approve a 7-month pilot on one line, ending in shadow mode. The model already clears accuracy, speed and demo; only Jindal's own strip can close false alarms, roll marks and edge cracks.